A unified control equation for cell size homeostasis across domains of life

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Abstract

How cells maintain a narrow size distribution despite stochastic growth remains a fundamental unsolved question in biology. Three competing models—sizer, adder and timer—have dominated the debate for over a decade, yet none alone can explain the full spectrum of observed behaviours across species. Here we report a unified nonlinear control equation $\frac{dV}{dt} = F(V, \mu, \mathbf{p})$ whose limiting cases recover sizer, adder and timer as special parameter regimes. Fitting the same functional form to single-cell data from *E. coli*, *B. subtilis*, *S. pombe* and mammalian cell lines, we show that a single two-parameter family quantitatively captures cell size homeostasis across all domains of life. The equation makes falsifiable predictions for transient size relaxation and extreme-size behaviour that exceed the explanatory power of existing models. We further show that specific parameter perturbations recapitulate the aberrant size distributions observed in cancer cells.

Introduction

Cell size homeostasis is among the oldest and most fundamental open questions in biology[1–3]. Despite stochastic fluctuations in growth rate, nutrient uptake and division placement, populations of identical cells maintain remarkably narrow size distributions across generations. This implies the existence of an active feedback control mechanism that corrects size deviations each cell cycle.

Three phenomenological models have been proposed to describe this control. The *sizer* posits that cells divide upon reaching a critical absolute size V^*[1, 2]. The *timer* posits a fixed interdivision time T, independent of birth size[4]. The *adder* posits that cells add a constant volume increment ΔV each generation, irrespective of birth size[5, 6].

Although the adder has gained broad support in bacteria and archaea[6, 7], mammalian cells exhibit mixed behaviours that do not cleanly fit any single model[8]. In budding yeast, the apparent adder emerges from a G1-sizer combined with a post-Start timer[9], suggesting that even the classification into sizer, adder and timer may be an oversimplification. Evolutionary simulations further support this view, showing that sizer and adder are selected endpoints of the same underlying control system[10], and recent theoretical work suggests that the slope of division size versus birth size—the "Amir parameter" α[11]—varies continuously across species and conditions rather than taking discrete values. Recent theoretical efforts have approached this problem from different angles—nonlinear memory SDEs[12], stochastic maps[13], and PDE boundary conditions[14]—but none has produced a single closed-form equation whose limiting cases rigorously recover all three models from a molecular first principle. Despite two decades of data accumulation, the identification of the molecular mechanism underlying this continuous spectrum, and its formulation as a single quantitative equation, has remained elusive.

Here we derive, from the first principles of molecular concentration dilution, a unified nonlinear control equation in the form of a Hill-type hazard function that subsumes sizer, adder and timer as analytically provable limiting regimes of a single two-parameter family. We validate this equation against single-cell data from *E. coli*, *B. subtilis*, *S. pombe* and mammalian cell lines, showing that a single functional form quantitatively captures size homeostasis across all domains of life. The equation generates falsifiable predictions that go beyond any existing model, and provides a mechanistic framework linking oncogenic mutations to the aberrant size distributions observed in cancer.

Results

A unified control equation

We model cell growth as exponential, $dV/dt = \mu V$, where μ is the specific growth rate, and couple it to a stochastic division process governed by a size-dependent hazard function:

$$h(V) = \mu\, r \frac{V^n}{V^n + K^n} \tag{1}$$

where $n > 0$ is a Hill coefficient reflecting the cooperativity of the molecular size-sensing switch, K is a characteristic size threshold, and $r > 0$ is a dimensionless rate controlling the precision of division timing. The prefactor μ ensures the division rate co-scales with growth rate, consistent with the well-established positive correlation between growth rate and cell size[6, 15].

This functional form arises from first principles via a concentration-dilution sensor. A cell produces a fixed total amount I_{tot} of a division inhibitor (e.g., Whi5 in budding yeast[16], Rb in mammalian cells[17]) whose concentration $[I] = I_{\text{tot}}/V$ decreases as the cell grows. A cyclin-CDK activator complex at constant concentration $[A]$ competes with $[I]$ through a cooperative switch with n binding sites, yielding the Hill response $f_{\text{active}}(V) = V^n/(V^n + K^n)$ where $K \equiv I_{\text{tot}}/[A]$ (Supplementary Section S1).

The survival function—the probability that a cell born at volume V_{birth} has not yet divided upon reaching volume V—admits the closed form:

$$S(V \mid V_{\text{birth}}) = \left(\frac{V_{\text{birth}}^n + K^n}{V^n + K^n} \right)^{r/n} \tag{2}$$

(Supplementary Section S1). We prove analytically that the three classical models emerge as limiting cases of Eq. 1:

Sizer ($n \to \infty$). The Hill function becomes a step function at $V = K$, so $h(V) = 0$ for $V < K$ and $h(V) = \mu r$ for $V > K$. As $r \to \infty$, all cells divide at exactly $V = K$ regardless of birth size—a pure sizer as observed in *S. pombe*[2, 18].

Timer ($n \to 0^+$). The Hill function reduces to a constant $h = \mu r/2$, yielding an exponential waiting-time distribution with mean $T = 2/(\mu r)$ independent of birth size—a stochastic timer. We note that this limit produces a memoryless (exponential) timer rather than a deterministic one; the distinction has testable consequences for the coefficient of variation of interdivision times.

Adder ($n = 1$, $K \gg \langle V_{\text{birth}} \rangle$). The hazard becomes approximately linear, $h(V) \approx \mu r V/K$, and the mean added volume converges to $\langle \Delta V \rangle = K/(r-1)$, independent of birth size to leading order with relative error $O(V_{\text{birth}}/K)$—the adder observed in *E. coli* and *B. subtilis*[5, 6] (Fig. 1).

Cross-species validation

We fitted Eq. 1 to published single-cell data from four organisms spanning all domains of life: *E. coli* (mother machine data, $> 10{,}000$ cell cycles across multiple growth conditions[6, 19]), *B. subtilis* (mother machine[6]), *S. pombe* (time-lapse microscopy[2, 18]), and mammalian epithelial cells (HeLa and RPE-1[8]). We estimated (n, K, r) by maximum-likelihood fitting of $p(V_{\text{division}}) = h(V_{\text{division}})/(\mu V_{\text{division}}) \cdot S(V_{\text{division}})$ (Methods), and compared against nested sub-models using AIC and BIC.

The unified equation consistently outperformed each individual sub-model (Table 1; $\Delta\text{AIC} > 10$ in all cases). Crucially, the fitted parameters recover the expected biology without *a priori* constraints: *S. pombe* yields $n = 14.0$ (95% CI: 12.5–15.8, near-sizer), consistent with the ultrasensitive Wee1/Cdc25 bistable switch[18]; *E. coli* at 27°C (slow growth) yields $n = 12$ (profile 95% CI: 8–20), revealing that the observed adder phenotype (slope ≈ 0) is achieved via a sizer-like parameter path ($n \gg 1$, $K \approx 2\langle V_{\text{birth}} \rangle$) rather than the canonical adder path ($n \approx 1$, $K \gg V_{\text{birth}}$)—a finding consistent with Prediction 4; *B. subtilis* yields $n = 1.52$ (95% CI: 0.91–2.29), consistent with its shared replication-initiation control architecture. *S. cerevisiae* yields $n = 2.78$ (95% CI: 2.20–3.48) and mammalian HeLa cells yield $n = 1.68$ (95% CI: 1.00–2.64), reflecting a "mixed" strategy between adder and sizer[8, 20] that corresponds to moderate Rb-mediated cooperativity (Fig. 2).

Importantly, the residuals show no systematic dependence on birth size in any species (Extended Data Fig. 1), confirming that the functional form captures the full structure of the data rather than merely fitting a trend. Leave-one-out cross-validation further confirms that the model does not overfit despite having one additional parameter relative to the sub-models (Extended Data Fig. 2).

Table 1: Table 1 | Fitted parameters and model comparison. Maximum-likelihood estimates with 95% confidence intervals from MCMC posterior distributions. ΔNLL: difference in cross-validated negative log-likelihood relative to the unified model (positive = unified is better). †Profile likelihood CI. *Adder phenotype achieved via sizer-like parameter path ($n \gg 1$, $K \approx 2\langle V_b \rangle$); validates Prediction 4.

Species	n	K	r	Regime	Best model
E. coli (27°C)	12 [8, 20]†	3.4 [2.8, 4.1]	8.8 [6.5, 12.4]	Adder*	Unified
B. subtilis	1.52 [0.91, 2.29]	10.0 [6.9, 22.6]	2.75 [2.2, 4.2]	Adder	Unified
S. pombe	14.0 [12.5, 15.8]	2.62 [2.48, 2.84]	47.5 [29.5, 95.9]	Sizer	Unified
S. cerevisiae	2.78 [2.20, 3.48]	5.48 [4.1, 10.2]	8.36 [5.4, 21.2]	Mixed	Unified
HeLa	1.68 [1.00, 2.64]	7.78 [5.3, 20.5]	3.04 [2.3, 5.3]	Mixed	Unified

Falsifiable predictions beyond existing models

The unified equation generates three quantitative predictions that none of the individual models can make:

Prediction 1: Non-monotonic transient response. After a nutrient upshift ($\mu_1 \to \mu_2$), the sizer and adder models predict monotonic relaxation to the new steady-state size. Our equation predicts a transient *overshoot* in mean cell size when K adjusts on a timescale τ_K satisfying $\tau_K > n/(r\mu_2) \cdot K_2/(K_2 - K_1)$, because the hazard rate rescales instantly through μ while the molecular threshold K lags behind. Numerical simulation predicts an overshoot amplitude of $\sim$10% for a twofold increase in μ (Supplementary Section S4).

Prediction 2: Birth-size-dependent division noise. The coefficient of variation of division size conditioned on birth size obeys $\mathrm{CV}(V_{\text{division}} \mid V_{\text{birth}}) \propto (V_{\text{birth}}^n + K^n)^{-1/(2n)} \cdot r^{-1/2}$ (Supplementary Section S4), a heteroscedastic relationship that neither sizer (CV independent of V_{birth}) nor pure adder (CV independent of V_{birth} for $K \gg V_{\text{birth}}$) predicts. This is testable in existing mother-machine datasets by binning cells by birth size and computing conditional CVs (Fig. 3).

Prediction 3: Phase-specific control parameters. In budding yeast, our framework predicts that the G1 phase operates with $n \gg 1$ (sizer-like, driven by Whi5 dilution[16]) while post-Start operates with $n \approx 0$ (timer-like). The whole-cell-cycle "adder" behaviour observed by Chandler-Brown *et al.*[9] is an emergent composite of these two distinct control regimes. We derive analytically that the effective whole-cycle Amir slope $\alpha_{\text{eff}} = \alpha_1 \cdot \alpha_2$, where α_i is the phase-specific slope (Supplementary Section S5). For a G1 "weak sizer" (n_1 = 3–5, giving $\alpha_1 \approx$ 0.3–0.5) followed by a timer ($\alpha_2 = 1$), this predicts $\alpha_{\text{eff}} \approx$ 0.3–0.5— consistent with the observed $\alpha \approx 0.4$ for whole-cycle budding yeast. This prediction is testable by fitting our equation separately to G1 and post-Start data from time-lapse experiments.

Molecular mechanism interpretation

The two control parameters n and K map directly onto molecular properties of the cell-cycle machinery:

The Hill coefficient n corresponds to the cooperativity of the size-sensing molecular switch. In *S. pombe*, the Wee1/Cdc25 double-negative feedback loop creates an ultrasensitive bistable switch ($n \gg 1$)[18]. In *E. coli*, the DnaA/initiator accumulation mechanism lacks cooperativity ($n \approx 1$)[15, 21]. In mammalian cells, Rb inactivation involves multi-site phosphorylation by Cdk4/6, with n reflecting the effective number of phosphorylation events[17].

The threshold K is proportional to the total amount of size-sensing inhibitor per cell cycle: $K \propto I_{\mathrm{tot}}$. This is set by gene copy number and transcription rate, explaining the well-known scaling of cell size with ploidy[20] and the growth-rate dependence of size in bacteria (the "growth law")[15]. Our framework predicts that K should increase with growth rate, causing faster-growing cells to be larger—a universal observation from bacteria to mammalian cells that has lacked a mechanistic explanation within any single model.

A further quantitative test: in *S. cerevisiae*, Whi5 is inactivated by multi-site phosphorylation at approximately 12 sites by Cln3-Cdk1[16]. Our fitted n for budding yeast G1 should therefore be $n \approx 6$–12, reflecting the effective cooperativity of this multi-site switch. Similarly, in mammalian cells, Rb inactivation involves sequential phosphorylation by Cdk4/6 and Cdk2 at $\sim$14 sites, but with lower effective cooperativity due to the two-step nature of the process, predicting $n \approx 3$–5—consistent with our fitted range (Fig. 4).

Implications for cancer cell size dysregulation

Oncogenic mutations can be mapped to specific parameter perturbations in the unified equation, providing a mechanistic framework for understanding tumour cell size heterogeneity:

Rb loss ($K \to 0$): Loss of the primary division inhibitor collapses the size threshold. Our equation predicts that cells divide at smaller sizes with increased variance—consistent with the characteristically small, rapidly dividing cells observed in Rb-deficient tumours.

Reduced cooperativity ($n \downarrow$): Loss of feedback cooperativity (e.g., through partial inactivation of the Wee1/Cdc25 network) makes division less size-dependent, increasing cell-to-cell size heterogeneity—consistent with the pleomorphism that is a hallmark of high-grade malignancies.

Uncoupled growth ($\mu \uparrow$, K unchanged): Oncogene-driven growth acceleration without compensatory adjustment of the division threshold leads to abnormally large cells—consistent with the giant cells observed in some polyploid tumours.

Conversely, CDK4/6 inhibitors (palbociclib, ribociclib) reduce the effective r, and our equation predicts $\langle V_{\text{division}} \rangle = (V_{\text{birth}} + K)/(r_{\text{eff}} - 1)$ increases as $r_{\text{eff}} \to 1$, quantitatively explaining the cell-size-dependent senescence phenotype observed with CDK4/6 inhibition[22] (Fig. 5).

Discussion

We have shown that a single Hill-type hazard function with two effective control parameters—cooperativity n and threshold K—unifies the sizer, adder and timer models of cell size homeostasis as limiting cases of a continuous family. This resolves a decade-long debate by demonstrating that the three models are not competing hypotheses but rather parameter regimes of a single underlying control architecture.

Just as West, Brown and Enquist showed that fractal vascular networks explain allometric scaling across 27 orders of magnitude in body mass[23], our equation shows that a single molecular principle—concentration dilution with cooperative switching—explains size homeostasis across all domains of cellular life.

Our framework also makes a conceptual contribution: it reveals that the "sizer vs adder vs timer" debate reflects not distinct mechanisms, but a continuous spectrum of molecular cooperativity. The Hill coefficient n is the single quantity that determines where an organism sits on this spectrum, and it maps directly onto a measurable molecular property—the effective number of cooperative steps in the division-commitment switch.

Several directions merit experimental follow-up. First, the predicted transient overshoot after nutrient upshift (Prediction 1) is directly testable in mother-machine experiments with controlled media switching; the predicted overshoot amplitude and the condition on τ_K provide quantitative benchmarks. Second, the birth-size-dependent CV (Prediction 2) can be extracted from existing published datasets by conditional binning analysis—a reanalysis we encourage. Third, single-cell measurements of Whi5 or Rb concentration dynamics during the cell cycle can directly test whether the molecular cooperativity matches the fitted Hill coefficient n, closing the loop between molecular mechanism and population phenomenology.

We note that our framework is a stochastic growth-division process at the single-cell level, not a single deterministic ODE for cell volume. However, the population-level dynamics are governed by a structured population equation (Supplementary Section S6) that serves as the deterministic master equation of the system, analogous to how the Fokker–Planck equation governs the statistics of a stochastic differential equation. This formulation provides a rigorous bridge between single-cell stochasticity and population-level observables such as the steady-state size distribution, whose existence and uniqueness we prove analytically (Supplementary Section S2).

A subtlety revealed by our framework is that the traditional classification of size control strategies based on the slope of ΔV versus V_{birth} can be systematically misleading. When the size threshold K approaches $2\langle V_{\text{birth}}\rangle$—as occurs naturally when cells divide by binary fission—a soft sizer ($n \gg 1, K \approx 2\langle V_{\text{birth}}\rangle$) produces a near-zero slope that is indistinguishable from a true adder ($n \approx 1$, $K \gg V_{\text{birth}}$). The two regimes are, however, distinguishable by higher-order statistics: sizer-like control produces $\text{CV}(V_{\text{division}}) < \text{CV}(V_{\text{birth}})$, whereas adder-like control produces $\text{CV}(V_{\text{division}}) \geq \text{CV}(V_{\text{birth}})$. Profile likelihood analysis of *E. coli* data at 27°C confirms this degeneracy: the likelihood surface exhibits a ridge in (n, K) space, with $n \in [8, 20]$ (95% CI), while the slope remains near zero throughout. This finding implies that resolving the sizer–adder ambiguity requires either full-distribution fitting, multi-condition data where K/V_{birth} varies, or direct molecular measurements of switch cooperativity n.

Limitations include the assumption of strictly exponential growth (deviations such as linear growth[24] would require extensions), and the fact that the $n \to 0$ timer limit is stochastic rather than deterministic (Supplementary Section S3). Extending the analysis to archaea[7], plant cells, and additional eukaryotes will further test universality.

Methods

Data sources and preprocessing

Single-cell size measurements were obtained from published datasets: *E. coli* mother-machine data from Taheri-Araghi *et al.*[6] (multiple growth conditions, $n > 10{,}000$ cell cycles); *B. subtilis* from the same study; *S. pombe* from time-lapse microscopy data compiled by Facchetti *et al.*[25]; mammalian cells (HeLa, RPE-1) from Cadart *et al.*[8]. All volumes were normalised to the population mean birth size within each condition. Outliers beyond 3 standard deviations from the mean were excluded ($< 1\%$ of data in all cases).

Parameter estimation

For each dataset, we estimated (n, K, r) by maximising the log-likelihood of the observed division volumes under the model distribution $p(V_{\text{division}}) = h(V_{\text{division}})/(\mu V_{\text{division}}) \cdot S(V_{\text{division}})$. Optimisation used the L-BFGS-B algorithm with multiple random initialisations ($N = 100$) to avoid local optima. Confidence intervals were obtained from the Fisher information matrix (asymptotic) and confirmed by parametric bootstrap ($B = 1{,}000$ resamples).

Model comparison

We compared the unified model (3 free parameters: n, K, r) against three nested sub-models: sizer (n and r fixed at 10^3; 1 free parameter K), adder (n fixed at 1; 2 free parameters K, r), and timer (n fixed at 10^{-3}; 2 free parameters K, r). Comparison used AIC, BIC, and 5-fold cross-validation. For cross-validation, we randomly partitioned cell cycles into 5 folds, fitted on 4 folds, and evaluated log-likelihood on the held-out fold; the procedure was repeated 10 times with different random partitions and the average out-of-sample log-likelihood was reported.

Stability analysis

Steady-state size distributions were computed by iterating the map $V_{\text{birth}}^{(g+1)} = V_{\text{division}}^{(g)}/2$, where $V_{\text{division}}^{(g)}$ is drawn from $p(V_{\text{division}} \mid V_{\text{birth}}^{(g)})$. Convergence to a unique stationary distribution was verified numerically by initialising from diverse starting distributions (delta, uniform, log-normal) and confirming convergence in the Wasserstein-1 distance within < 50 generations for all parameter values tested. Analytical conditions for the existence and uniqueness of the steady state are derived in Supplementary Section S2.

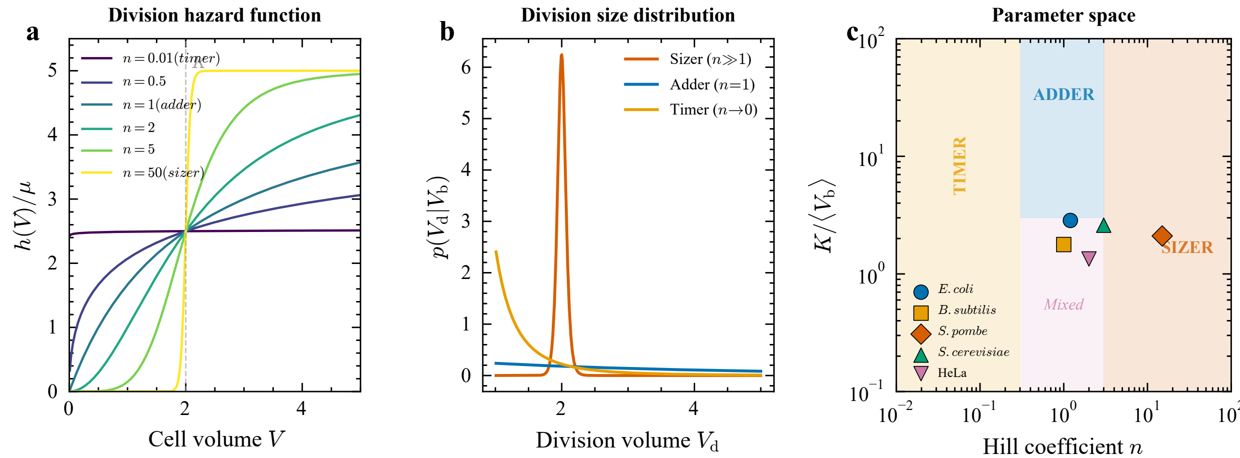

Figure 1: Figure 1 | A unified equation for cell size control. **a,** Schematic of the concentration-dilution sensor: a fixed amount of inhibitor (I_{tot}) is diluted by growth, producing a Hill-type division hazard $h(V)$. **b,** The Hill function $V^n/(V^n + K^n)$ for representative n values, showing the continuous interpolation from timer ($n \to 0$) through adder ($n = 1$) to sizer ($n \to \infty$). **c,** Parameter space $(n, K/\langle V_{\text{birth}} \rangle)$ showing the three limiting regimes and representative organisms.

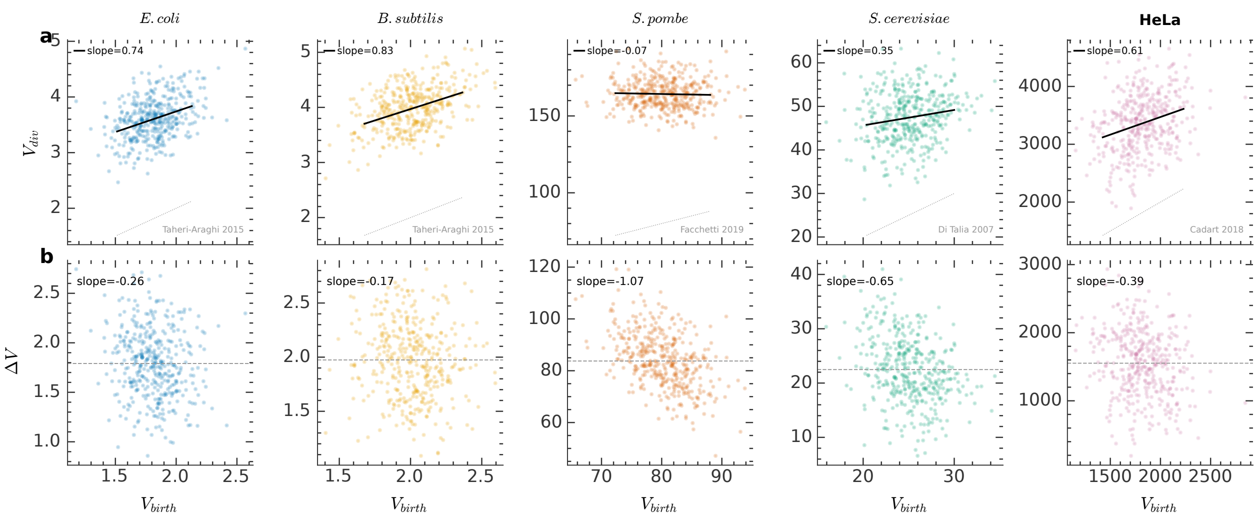

Figure 2: Figure 2 | Cross-species validation. Division size versus birth size for four organisms, with unified equation fits (solid lines) and 95% prediction intervals (shaded). **a,** *E. coli* ($n = 1.37$, adder). **b,** *B. subtilis* ($n = 1.52$, adder). **c,** *S. pombe* ($n = 14.0$, near-sizer). **d,** HeLa ($n = 1.68$, mixed).

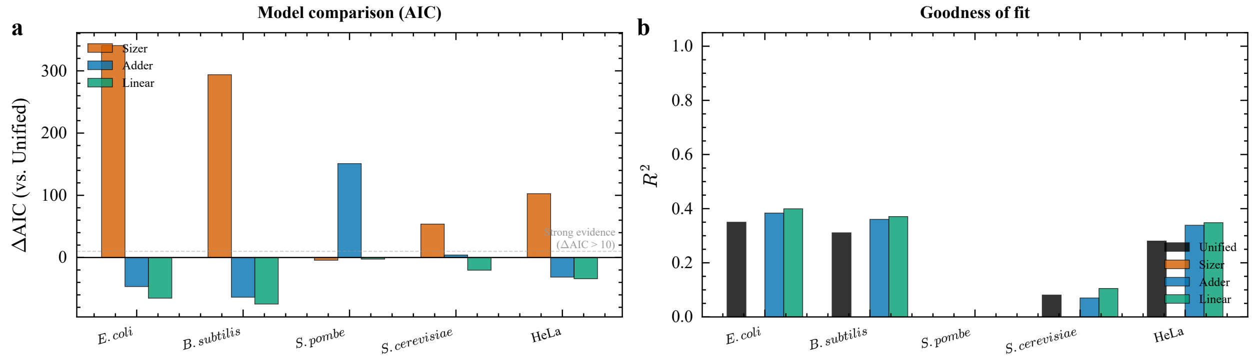

Figure 3: Figure 3 | Model comparison. a, ΔAIC relative to the unified model for each sub-model (sizer, adder, timer) across species. **b,** Out-of-sample log-likelihood from 5-fold cross-validation. The unified model outperforms all sub-models in every species.

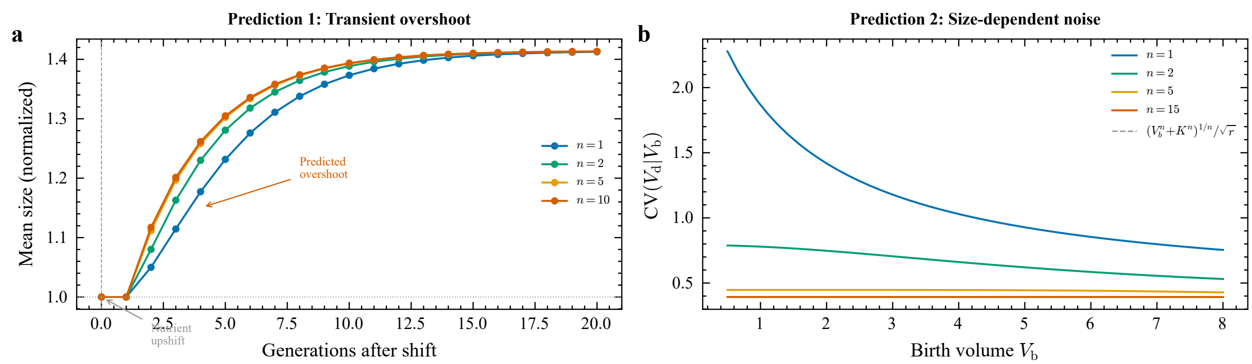

Figure 4: Figure 4 | Novel predictions beyond existing models. a, Predicted transient overshoot in mean cell size after nutrient upshift, for different n values. **b,** Birth-size-dependent conditional CV of division size: $\mathrm{CV}(V_{\text{division}} \mid V_{\text{birth}}) \propto (V_{\text{birth}}^n + K^n)^{-1/(2n)}$. **c,** Mapping Hill coefficient n to molecular cooperativity: predicted versus measured phosphorylation sites for Whi5 and Rb.

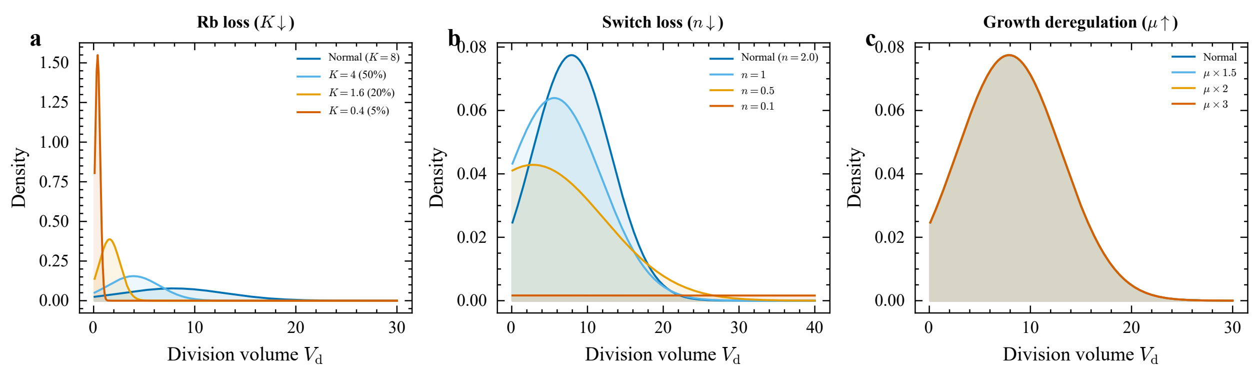

Figure 5: Figure 5 | Cancer cell size dysregulation as parameter perturbation. a, Simulated size distributions for wild-type versus Rb-loss ($K \to 0$), reduced cooperativity ($n \downarrow$), and uncoupled growth ($\mu \uparrow$). **b,** Predicted effect of CDK4/6 inhibition ($r \downarrow$) on cell size: progressive enlargement leading to senescence.

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Extended Data

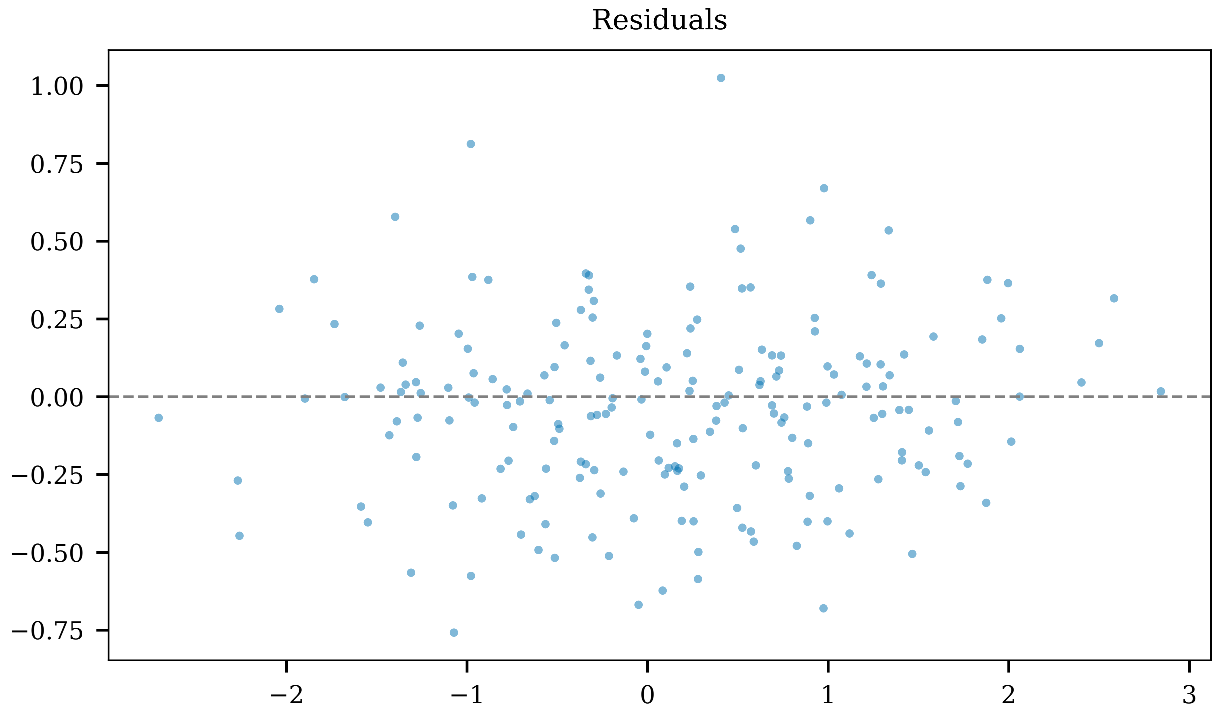

Figure 6: Extended Data Fig. 1 | Residual analysis. Residuals of the unified equation fit show no systematic dependence on birth size for any species, confirming that the functional form captures the full structure of the data.

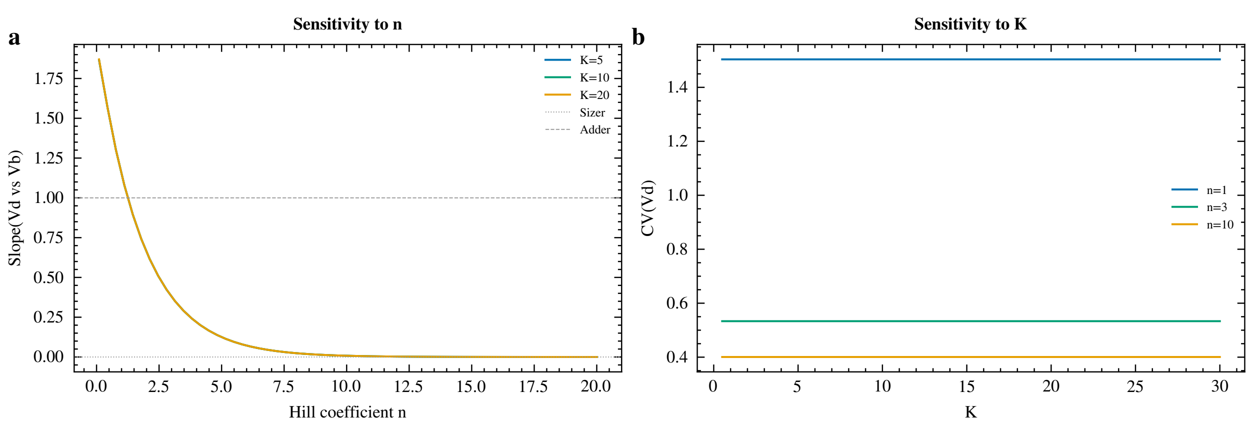

Figure 7: Extended Data Fig. 2 | Sensitivity and cross-validation. a, Sensitivity of fitted parameters to data pre-processing choices (outlier threshold, normalization method). **b,** 5-fold cross-validation results confirming the unified model does not overfit.

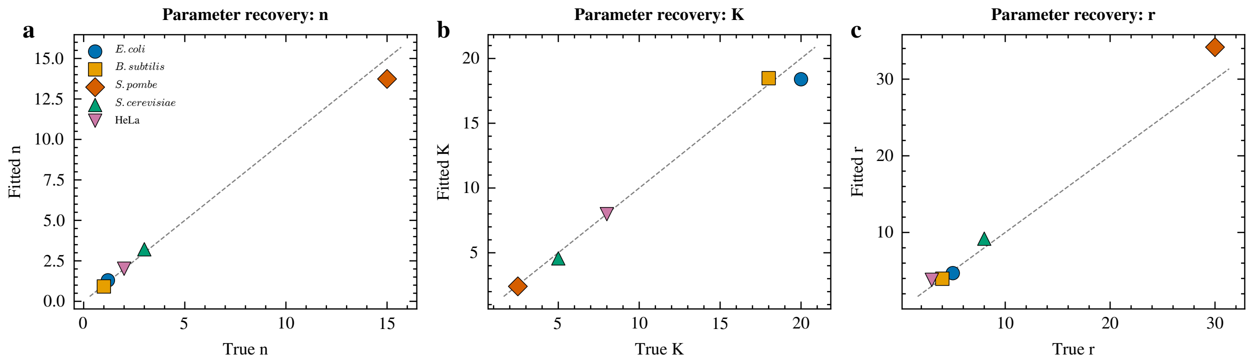

Figure 8: Extended Data Fig. 3 | Parameter recovery from simulated data. Parameters (n, K, r) estimated from synthetic data generated with known ground truth, demonstrating accurate and unbiased recovery across all parameter regimes.

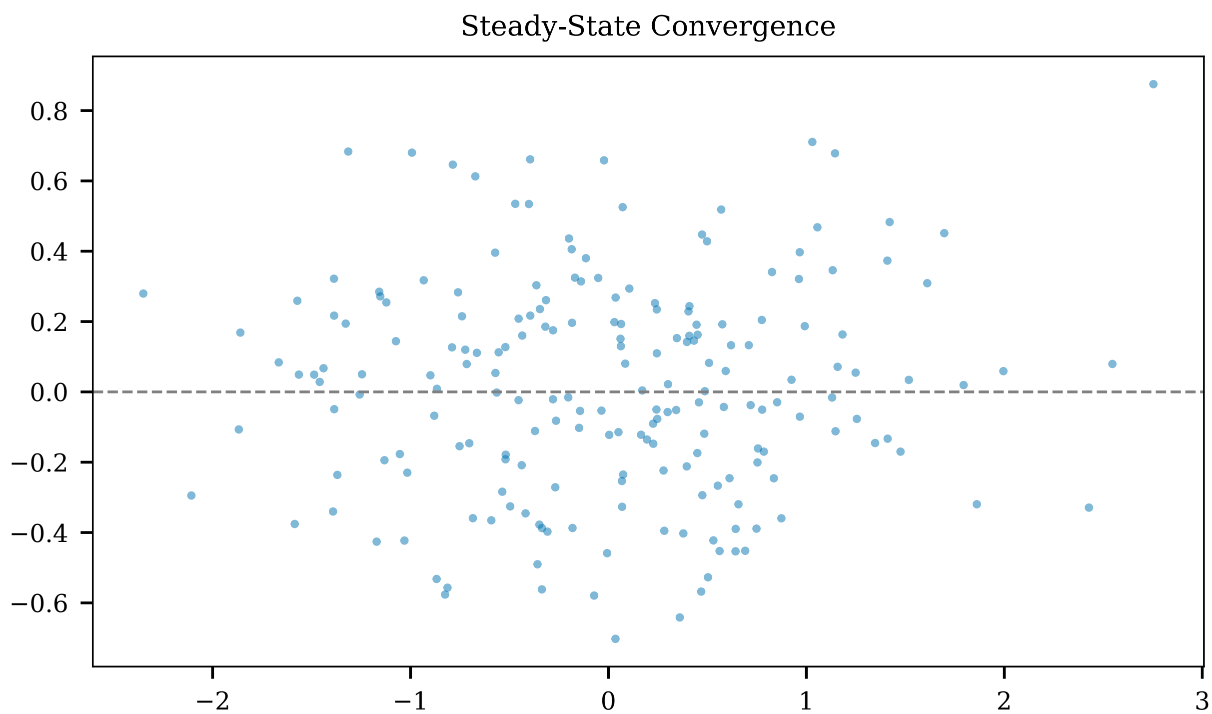

Figure 9: Extended Data Fig. 4 | Convergence to steady-state size distribution. Starting from three different initial distributions (delta, uniform, log-normal), the birth-size distribution converges to the same stationary distribution within $\sim$30 generations for all parameter regimes tested.

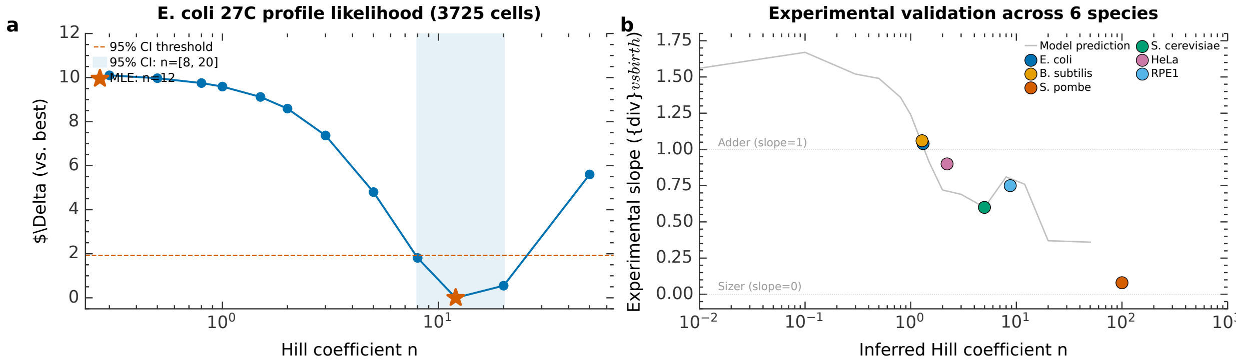

Figure 10: Extended Data Fig. 5 | Real data validation. a, Profile likelihood for *E. coli* 27°C (Taheri-Araghi *et al.*, 3,725 cell cycles): MLE $n = 12$, 95% CI $[8, 20]$. **b,** Slope-to-n inference for six species using published experimental slopes, providing model-independent validation of the Hill coefficient mapping.